

Assignment 2AD, 2025

DD2434 Machine Learning, Advanced Course

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D Section

Question D.1

I chose the dot product similarity as similarity function

$$Y = WX \quad Y \in R^{d \times n}$$

d represents the dimensionality of the data while n is the number of data in the data set and k is the new dimensionality of the data.

We know that W is orthonormal ($W^T W = I$)

$$S = Y^T Y = (WX)^T (WX) = X^T (W^T W) X = X^T X$$

S matrix is the Gram matrix.

Why this choice is meaningful:

- Interpretation → the dot product measures the alignment between two MEPs' voting patterns. Higher values indicate greater agreement.
- Domain appropriateness → in voting behavior analysis, we care about agreement on specific votes. The dot product naturally captures co-voting patterns (when both MEPs vote the same way, it contributes positively to similarity)
- Simplicity

Transformation steps required for our similarity computation:

- Encoding of the EPG column
 - 0 (not an MEP) → exclude from computation
 - 1 (for) → +1
 - 2 (against) → -1

- 3 (abstention) $\rightarrow 0$
- 4 (absent) $\rightarrow 0$
- 5 (did not vote) $\rightarrow 0$
- 6 (motivated) $\rightarrow 0$

"For" and "Against" are opposite positions (+1, -1)

- Filtering $\rightarrow$ MEPs with a very low participation rate are ignored
- Missing Data $\rightarrow$ for each pair of MEPs, only include votes where BOTH MEPs were present and voted. This ensures we're measuring similarity based on actual voting decisions, not absence patterns.
- Center $\rightarrow$ subtract each MEP's mean vote in order to remove the baseline voting tendency of each MEP.
- Normalize $\rightarrow$ divide by L2 norm in order to make similarity values comparable across MEP pairs with different participation rates.

I selected MEPs from both the EP8 (2014-2019) and EP9 (2019-2022) sessions, applying a minimum participation threshold of 30%. This resulted in a subset of 1,520 MEPs who voted on 10,251 recorded votes, yielding approximately 2.3 million pairwise similarity comparisons. Reasons:

- combining two consecutive parliamentary sessions provides a broader temporal perspective, allowing the analysis to capture both stable political alignments and potential shifts in voting behavior across time.
- the 30% participation threshold ensures that similarity estimates are based on a substantial number of co-voted issue
- the computed similarity matrix shows expected properties: perfect self-similarity (diagonal ≈ 1.0), symmetry, and a reasonable distribution of similarities (mean = 0.14, std = 0.27) with both high agreement (max = 0.998) and strong opposition (min = -0.42) represented.

Question D.2

We want to find the latent matrix $X \in R^{k \times n}$ such that $S = X^T X$.

The similarity matrix S is symmetric positive semi-definite, so its eigen-decomposition is

$$S = U \Lambda U^{-1} = U \Lambda U^T = (\Lambda^{0.5} U^T)^T (\Lambda^{0.5} U^T)$$

The non negative eigenvalues of S are sorted in descending order in the diagonal of Λ , we can take the $k \times n$ matrix X to be

$$X = I_{k \times n} \Lambda^{0.5} U^T$$

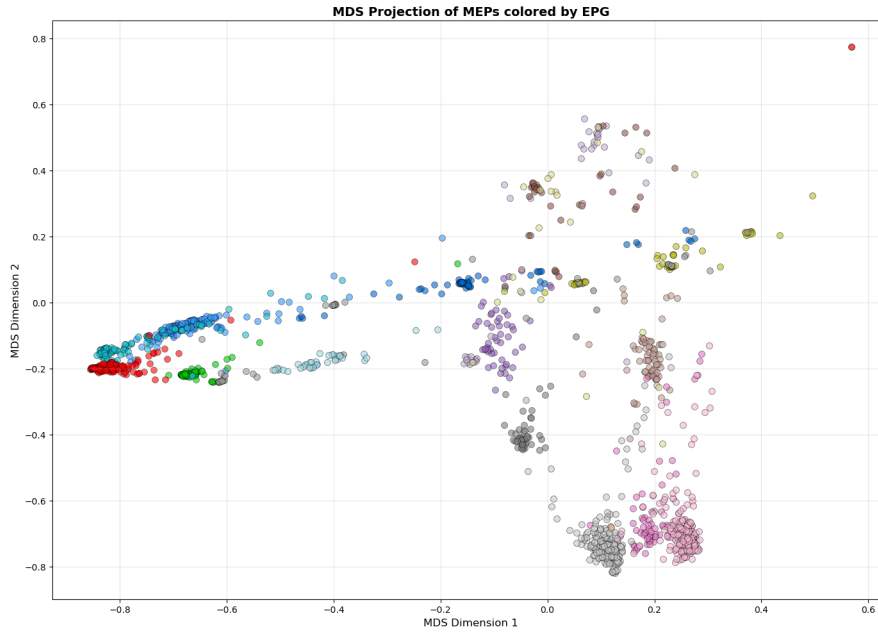


Figure 1: MEPs colored by Political Group (EPG)



Figure 2: MEPs colored by Country

In my model , I assumed $k = 2$

The MDS visualization reveals that political group (EPG) affiliation is the dominant

factor explaining MEP voting patterns. When colored by EPG, the plot shows clear, well-separated clusters following a classic left-right political spectrum with a distinctive horseshoe pattern. In contrast, when colored by country, the visualization shows extensive overlap with no coherent national clusters.

This finding indicates that MEPs vote primarily along ideological lines rather than national interests. The strong cohesion within European political groups (e.g., EPP, S&D) suggests that party discipline and shared political values transcend national boundaries in the European Parliament.

The horseshoe pattern is particularly noteworthy: the left-center-right spectrum is clearly visible, with centrist parties (EPP, Renew) at the top of the arc, left-wing parties (S&D, Greens, GUE/NGL) on the left side, and right-wing/Eurosceptic parties (ECR, ID) on the right. The fact that political extremes appear closer to each other than to the center likely reflects their shared Eurosceptic positions despite opposite economic ideologies.

Question D.3

In this section, we aim to conduct a more nuanced analysis of the MEP voting dataset by addressing the following research questions:

Do Members of the European Parliament (MEPs) tend to vote along party lines when a vote is sponsored by their own political group? Furthermore, which political groups exhibit systematic voting alliances?

Hypothesis:

- Party loyalty → MEPs will vote overwhelmingly in favor of proposals sponsored by their own political group (high in-group support)
- Coalition patterns → certain political groups will consistently support each other's proposals, revealing stable legislative alliances (e.g., EPP-S&D "grand coalition", or Greens-S&D left-wing cooperation)
- Opposition patterns → ideologically distant groups will systematically vote against each other's proposals (e.g., far-left vs far-right, or Eurosceptics vs federalists)

I used the support matrix with the dataset composed by EP9_RCVs_2022_06_22 and EP9_Voted_docs.

Support matrix is a quantitative tool that measures how political groups vote on each other's legislative proposals. It is constructed as follows:

- Rows → Political groups acting as sponsors (proposing votes/amendments)
- Columns → Political groups acting as voters (voting on those proposals)
- Cell values → support rate computed as $SupportRate = \frac{\text{Number of "For" votes by voter group}}{\text{Total votes cast by voter group}}$

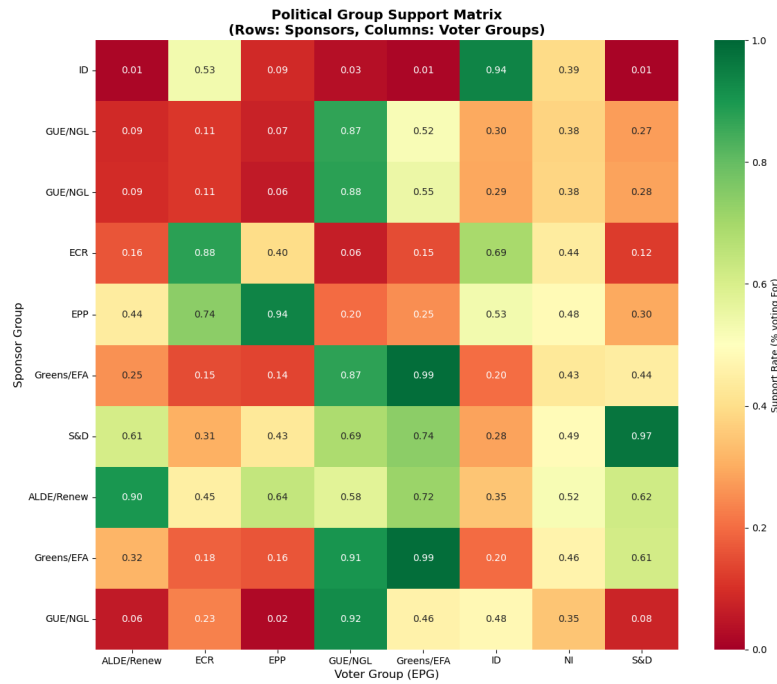


Figure 3: Support Matrix

Metodology:

- Step 1 → Data Integration
- Step 2 → Group Name Standardization: For each political group acting as sponsor: political groups have multiple naming variations across files (e.g., "EUL/NGL", "The Left", and "Confederal Group of the European United Left - Nordic Green Left" all refer to GUE/NGL). We created a mapping dictionary to standardize all group names to short, consistent labels.
- Step 3 → Vote Matching, for each political group acting as sponsor:
 - Identified all votes in the Voted_docs file where that group is listed as Author

- Matched these votes to the corresponding columns in the RCV file using position-based indexing
- Extracted the voting behavior of all MEPs on those specific votes
- Step 4 → Support Rate Calculation, for each (sponsor, voter) pair
 - Filtered votes to only those sponsored by the sponsor group
 - Extracted how MEPs from the voter group voted on those proposals
 - Applied the formula $\text{Support Rate} = \frac{(\# \text{ of For votes})}{(\# \text{ of For} + \text{Against} + \text{Abstain})}$

Key findings:

- **Very high in-group loyalty** across all groups (87-99%), with Greens/EFA (99%) and S&D (97%) showing the strongest discipline. This indicates that European political groups function as cohesive voting blocs with strong party discipline.
- **Three distinct voting blocks**
 - Left alliance (S&D , Greens/EFA , GUE/NGL): Mutual support ranging from 61-99%
 - Center-right coalition (EPP , ECR): Moderate cooperation (40-74%)
 - Liberal bridge (ALDE/Renew): Supports both left and right groups moderately (62-72%), acting as a swing voter
- **Political isolation of the far-right** The ID group (Identity and Democracy) receives near-zero support from all other groups (1-9%), while maintaining high internal loyalty (94%)
- **Grand Coalition** EPP and S&D, traditionally the two largest groups, show asymmetric support (EPP→S&D: 30%, S&D→EPP: 43%). While they cooperate on major legislation, the moderate support levels confirm that ideological differences remain significant and coalition formation requires negotiation rather than automatic agreement.
- **Strong left-wing cooperation** The highest cross-group support rates appear among left-wing groups, particularly Greens/EFA supporting GUE/NGL proposals (91-99%), indicating that the left bloc votes together more cohesively than the right.

MDS showed that groups vote similarly while Support matrix shows who leads and who follows (e.g., Greens support S&D proposals 74%, but S&D supports Greens 61% - slightly asymmetric). This makes the support matrix analysis particularly valuable for understanding legislative effectiveness and coalition formation in the European Parliament rather than just having an overall ideological structure with MDS.

C Section

Question C.1

Considering a dataset matrix $Y \in R^{d \times n}$, its SVD $Y = U\Sigma V^T$ and the similarity matrix $S = Y^T Y$, we want to prove that $S = V\Sigma^2 V^T$

$$\begin{aligned} S &= Y^T Y = (U\Sigma V^T)^T (U\Sigma V^T) \\ &= V\Sigma^T U^T U \Sigma V^T \end{aligned}$$

Since U is known to be orthonormal from the SVD definition

$$U^T U = I$$

$$S = V\Sigma^T \Sigma V^T$$

Σ is a diagonal matrix with no-negative entries

$$\Sigma^T = \Sigma \quad \Sigma^T \Sigma = \Sigma^2$$

We finally obtain

$$S = V\Sigma^2 V^T$$

Question C.2

We want to prove that the two embeddings (MDS and PCA) differ only by an orthogonal transform such that $X_{PCA} = R X_{MDS}$ and that for Euclidean data MDS and PCA have the yield to the same embedding (up to rotation)

Theoretical concepts:

- MDS embedding

$$S = U\Lambda U^{-1} = U\Lambda U^T$$

$$X_{MDS} = I_{k \times n} \Lambda^{0.5} U^T = \Lambda_k^{0.5} U_k^T$$

Λ_k is the diagonal matrix of top k eigenvalues while U_k represents the first k columns of U (top k eigenvectors)

- PCA embedding

$$Y = U^{SVD} \Sigma V^T$$

$$X_{PCA} = (U_k^{SVD})^T Y$$

Knowing that $S = Y^T Y = V\Sigma^2 V^T$ and $S = U\Lambda U^T$

- Eigenvectors of S are the right singular vectors V of Y
- Eigenvalues of S are the square singular values $\lambda_i = \sigma_i^2$

$$U = V \quad \Lambda = \Sigma^2$$

We can rewrite both X_{MDS} and X_{PCA} to find a common definition:

- $X_{MDS} = \Sigma_k V_k^T$
Replacing $U = V$ $\Lambda = \Sigma^2$ into its original definition $X_{MDS} = \Lambda_k^{0.5} U_k^T$
- $X_{PCA} = (U_k^{SVD})^T Y = (U_k^{SVD})^T (U_{SVD} \Sigma V^T) = \Sigma_k V_k^T$
We have obtained the final form by using this expression $(U_k^{SVD})^T U^{SVD} = [I_k | 0]$

We can compare the two expressions and see that:

$$X_{PCA} = \Sigma_k V_k^T \quad X_{MDS} = \Sigma_k V_k^T$$

$$X_{PCA} = I_k X_{MDS}$$

Where I_k is the identity matrix $R^{k \times k}$ which is orthogonal. We have proved that:

$$R = I_k$$

For Euclidean data (zero centered), classical MDS yields exactly the same embedding as PCA (they differ by an orthogonal transformation with $R = I_k$ which is the identity rotation)

In simpler words: MDS works on distances while PCA works on variances. They only align when the data is centered.

Question C.3

We want to show that the variance of the dataset $Var(Y) = \sum_{y \in Y} \|y - \bar{y}\|_2^2$ can be expressed as $Var(Y) = \sum_{i=1}^d \sigma_i^2$.

We know that the data is zero-centered ($\bar{y} = 0$)

$$\begin{aligned} Var(y) &= \sum_{y \in Y} \|y\|_2^2 \\ &= \sum_{i=1}^n \|y_i\|_2^2 \\ &= \sum_{i=1}^n \sum_{j=1}^d y_{ij}^2 \\ &= \|Y\|_F^2 \end{aligned}$$

$\|Y\|_F^2$ is the Frobenius Norm.

We want to show that:

$$\|Y\|_F^2 = \sum_{i=1}^d \sigma_i^2$$

We can start noticing that the Frobenius norm can be expressed using the trace

$$\begin{aligned}
\text{Trace}(Y^T Y) &= \sum_{j=1}^n (Y^T Y)_{jj} \\
&= \sum_{j=1}^n \sum_{i=1}^d Y_{ji}^T Y_{ij} \\
&= \sum_{j=1}^n \sum_{i=1}^d Y_{ij} Y_{ij} \\
&= \sum_{j=1}^n \sum_{i=1}^d Y_{ij}^2 \\
&= \|Y\|_F^2
\end{aligned}$$

We can prove that multiplying with an orthonormal matrix ($U^T U = I$ or $V^T V = I$) does not change the Frobenius norm:

$$\begin{aligned}
\|UY\|_F^2 &= \text{tr}((UY)^T (UY)) & \|YV^T\|_F^2 &= \text{tr}((YV^T)^T (YV^T)) \\
&= \text{tr}(Y^T U^T U Y) & &= \text{tr}(V Y^T Y V^T) \\
&= \text{tr}(Y^T Y) & &= \text{tr}(Y^T Y V^T V) \\
&= \|Y\|_F^2 & &= \text{tr}(Y^T Y) \\
& & &= \|Y\|_F^2
\end{aligned}$$

Knowing these properties and the SVD $Y = U \Sigma V^T$, we can finally show that:

$$\begin{aligned}
\|Y\|_F^2 &= \|U \Sigma V^T\|_F^2 \\
&= \|\Sigma V^T\|_F^2 \\
&= \|\Sigma\|_F^2 \\
&= \sum_{i=1}^n \sigma_i^2
\end{aligned}$$

Question C.4

We want to show that the variance of the projected data X is given by $\text{Var}(X) = \sum_{i=1}^k \sigma_i^2$

$$\begin{aligned}
X &= W^T Y \\
W &\in R^{d,k} \quad X \in R^{k,n} \quad Y \in R^{d,n}
\end{aligned}$$

$$\begin{aligned}
Var(X) &= \sum_{i=1}^n \|x_i - \bar{x}_i\|_2^2 \\
&= \sum_{i=1}^n \|x_i\|_2^2
\end{aligned}$$

Since Y is zero centered ($\bar{y} = 0$), the projected data is also zero centered:

$$\begin{aligned}
\bar{x} &= \frac{1}{n} \sum_{j=1}^n x_j \\
&= \frac{1}{n} \sum_{j=1}^n W^T y_j \\
&= W^T \bar{y} \\
&= 0
\end{aligned}$$

The proof is very similar to the previous exercise , we will exploit the trace its cyclic property.

$$\begin{aligned}
Var(X) &= \|X\|_F^2 \\
&= \|W^T Y\|_F^2 \\
&= \text{Trace}((W^T Y)^T (W^T Y)) \\
&= \text{Trace}(Y^T W W^T Y) \\
&= \text{Trace}(W W^T Y Y^T) \\
&= \text{Trace}(W W^T U \Sigma V^T V \Sigma^T U^T) \\
&= \text{Trace}(W W^T U \Sigma^2 U^T)
\end{aligned}$$

We know that $W = U_k$ from the slide of the Module 7 (PCA theory)

$$\begin{aligned}
Var(X) &= \text{Trace}(U_k U_k^T U \Sigma^2 U^T) \\
&= \text{Trace}(U_k^T U \Sigma^2 U^T U_k)
\end{aligned}$$

We know that:

$$U_k^T U = (I_k | 0) \quad U^T U_k = \begin{pmatrix} I_k \\ 0 \end{pmatrix}$$

It becomes:

$$\begin{aligned}
Var(X) &= \text{Trace} \left((I_k | 0) \Sigma^2 \begin{pmatrix} I_k \\ 0 \end{pmatrix} \right) \\
&= \sum_{i=1}^k \sigma_i^2
\end{aligned}$$

Question C.5

We consider the residual points $Z = z_1, \dots, z_n$ where $z_i = y_i - WW^T y_i$ and we want to show that its variance is given by $Var(Z) = \sum_{i=1}^d \sigma_i^2$

We start from the definition of variance

$$\begin{aligned} Var(Z) &= \sum_{i=1}^n \|z_i - \bar{z}\|_2^2 \\ &= \sum_{i=1}^n \|z_i\|_2^2 \\ &= \|Y - WW^T Y\|_F^2 \end{aligned}$$

The residual points are zero centered, below we have the proof for that.

$$\begin{aligned} \bar{z} &= \frac{1}{n} \sum_{i=1}^n y_i - WW^T y_i \\ &= \frac{1}{n} \sum_{i=1}^n y_i - WW^T y_i \\ &= \frac{1}{n} \sum_{i=1}^n y_i - \frac{1}{n} \sum_{i=1}^n WW^T y_i \\ &= \bar{y} - WW^T \bar{y} \\ &= 0 \end{aligned}$$

We will use the trace, its cyclic and linearity property for the proof, it is also very important to remember that $W^T W = I_k$

$$\begin{aligned} Var(Z) &= \|Y - WW^T Y\|_F^2 \\ &= \text{Trace}((Y - WW^T Y)^T (Y - WW^T Y)) \\ &= \text{Trace}(Y^T Y - Y^T WW^T Y - Y^T WW^T Y + Y^T WW^T WW^T Y) \\ &= \text{Trace}(Y^T Y - 2Y^T WW^T Y + Y^T WW^T WW^T Y) \\ &= \text{Trace}(Y^T Y - 2Y^T WW^T Y + Y^T WW^T Y) \\ &= \text{Trace}(Y^T Y - Y^T WW^T Y) \\ &= \text{Trace}(Y^T Y) - \text{Trace}(Y^T WW^T Y) \\ &= \text{Trace}(Y^T Y) - \text{Trace}(WW^T Y Y^T) \end{aligned}$$

Both of them have already been seen in the previous exercises, in particular:

- $\text{Trace}(Y^T Y)$ has been discussed in the Question C.3 where we proved that

$$\text{Trace}(Y^T Y) = \sum_{i=1}^d \sigma_i^2$$

- $\text{Trace}(WW^TYY^T)$ has been discussed in the Question C.4 where we proved that

$$\text{Trace}(WW^TYY^T) = \sum_{i=1}^k \sigma_i^2$$

The final expression becomes:

$$\begin{aligned} \text{Var}(Z) &= \text{Trace}(Y^TY) - \text{Trace}(WW^TYY^T) \\ &= \sum_{i=1}^d \sigma_i^2 - \sum_{i=1}^k \sigma_i^2 \\ &= \sum_{i=k+1}^d \sigma_i^2 \end{aligned}$$

We can conclude our discussion saying that:

$$\text{Var}(Y) = \text{Var}(X) + \text{Var}(Z)$$

$$\text{Var}(Y) = \sum_{i=1}^k \sigma_i^2 + \sum_{i=k+1}^d \sigma_i^2$$

Where $\text{Var}(Y)$ is the total variance and it can be computed by taking into account the variance explained by PCA ($\text{Var}(X)$) and the residual variance ($\text{Var}(Z)$)

B Section

Question B.1

D is the degree matrix, a diagonal matrix where:

- $D_{ii} = \sum_j A_{ij}$ which is the degree of node i (the number of edges connected to it)
- $D_{ij} = 0$ for all $i \neq j$ (off-diagonal entries are zero)

P is the transition probability matrix for a random walk on the graph:

$$P_{ij} = \frac{A_{ij}}{D_{ii}} = \frac{A_{ij}}{\text{degree of node i}}$$

P_{ij} is the probability of moving from node i to node j in one step if you randomly choose one of node i's neighbors.

The similarity measure S is defined using the variable k:

$$S_{ij} = \sum_{k=1}^{\infty} \alpha^k P_{ij}^k$$

k is the number of steps (or path length) in a random walk:

- $k=1$: one-step paths (direct neighbors)
- $k=2$: two-step paths (neighbors of neighbors)
- $k=3$: three-step paths, and so on...

Let's go deeper into the definition of the similarity measure:

- P_{ij}^k is the probability of reaching node j from node i in exactly k steps via a random walk. A large value indicates many k -step paths exist between i and j .
- α^k is the decay factor: it gives exponentially decreasing weight to longer paths, ensuring direct connections (small k) contribute more than distant connections (large k)
 - The decay parameter balances local vs. global structure: smaller α emphasizes direct connections, while larger α considers longer-range connections.
 - The decay parameter ensures we don't count very long, indirect paths too heavily (nearby structure matters more than distant connections).
 - The condition on the decay parameter ($0 < \alpha < 1$) allows the infinite sum to converge to a finite value
- $\sum_{k=1}^{\infty}$ sums the contributions from all path lengths, combining information from direct connections, common neighbors, and broader network structure.

S captures the idea that two nodes are similar if they are connected through many paths of various lengths.

Question B.2

For all $i, j \in V$, we want to prove that

- $P_{ij} \leq 1$

$$P = D^{-1}A$$

$$P_{ij} = \frac{A_{ij}}{D_{ii}} = \frac{A_{ij}}{\sum_j A_{ij}}$$

Because A_{ij} is either 0 or 1, and D_{ii} is the sum of all entries in row i , it follows that $A_{ij} \leq D_{ii}$. Consequently, the ratio is a binary value divided by a positive integer, which implies $0 \leq P_{ij} \leq 1$.

- $S_{ij} < +\infty$

$$S_{ij} = \sum_{k=1}^{\infty} \alpha^k P_{ij}^k$$

$$S = \sum_{k=1}^{\infty} \alpha^k P^k = \sum_{k=0}^{\infty} (\alpha P)^k - I$$

It is a geometric series which will converge to a finite value if $-1 < \alpha P < 1$: we know that $P_{ij} < 1$ and by definition of the problem we can assume $0 < \alpha < 1$.

$$S = (I - \alpha P)^{-1} - I$$

Since the result is a well-defined finite matrix, every entry S_{ij} must be finite ($S_{ij} < +\infty$).

Question B.3

We want to show that S can be computed efficiently using a matrix inversion operation.

$$\begin{aligned} S &= \sum_{k=1}^{\infty} \alpha^k P^k \\ &= \sum_{k=0}^{\infty} (\alpha P)^k - I \\ &= (I - \alpha P)^{-1} - I \end{aligned}$$

Complexity:

- Compute $I - \alpha P$ is $O(n^2)$
- Invert $I - \alpha P$ is $O(n^3)$

Total complexity is then $O(n^3)$ with $n = |V|$

A Section

Multidimensional Scaling (MDS) is a dimensionality reduction technique that aims to embed high-dimensional data into a lower-dimensional space while preserving pairwise distances. Given a distance matrix $\mathbf{D} \in \mathbb{R}^{n \times n}$ containing all pairwise distances d_{ij} between n data points, the goal is to find a k -dimensional representation $\mathbf{x}_i \in \mathbb{R}^k$ for each point i , where typically $k \ll n$. These vectors can be arranged into a configuration matrix $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_n] \in \mathbb{R}^{k \times n}$.

The quality of the embedding is measured by the stress function:

$$E(\mathbf{X}) = \sum_{i < j} w_{ij} (d_{ij} - \|\mathbf{x}_i - \mathbf{x}_j\|)^2, \quad (1)$$

where w_{ij} are weights that can either be specified as input or computed from the distances themselves. This objective quantifies the discrepancy between the original distances and the Euclidean distances in the embedded space.

Since the stress function does not admit a closed-form solution, numerical optimization methods such as gradient descent are employed to find the configuration $\mathbf{X}$ that minimizes $E(\mathbf{X})$.

Question A.1

We want to show that the gradient of E wrt a point x_i is

$$\nabla_{x_i} E = 2 \sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|} \right) (x_i - x_j)$$

We start from the definition of E :

$$E(X) = \sum_{i < j} w_{ij} (d_{ij} - \|x_i - x_j\|)^2$$

We apply the gradient wrt x_i and we used the linearity and chain rule property of it.

$$\begin{aligned} \nabla_{x_i} E(x) &= \nabla_{x_i} \left(\sum_{i < j} w_{ij} (d_{ij} - \|x_i - x_j\|)^2 \right) \\ &= \nabla_{x_i} \left(\frac{1}{2} \sum_{k=1}^n \sum_{j \neq k} w_{kj} (d_{kj} - \|x_k - x_j\|)^2 \right) w \end{aligned}$$

Here, the factor of $\frac{1}{2}$ accounts for counting each pair twice: once as (k,j) and once as (j,k) . When we take $\nabla_{\mathbf{x}_i}$ we need to find which terms in the double sum depend on x_i , it appears in:

- The outer sum when $k=i$: terms of the form $\|\mathbf{x}_i - \mathbf{x}_j\|$ for all $j \neq i$
- The inner sum when $j=i$: terms of the form $\|\mathbf{x}_k - \mathbf{x}_i\|$ for all $k \neq i$

The expression becomes

$$\nabla_{\mathbf{x}_i} E = \frac{1}{2} \left[\sum_{j \neq i} \nabla_{\mathbf{x}_i} (w_{ij} (d_{ij} - \|\mathbf{x}_i - \mathbf{x}_j\|)^2) + \sum_{k \neq i} \nabla_{\mathbf{x}_i} (w_{ki} (d_{ki} - \|\mathbf{x}_k - \mathbf{x}_i\|)^2) \right]$$

We can observe that the distances (d) and the weights (w) are symmetric and both the sums have the same exact form, it becomes:

$$\begin{aligned} \nabla_{x_i} E(x) &= \sum_{i \neq j} w_{ij} \nabla_{x_i} \left((d_{ij} - \|x_i - x_j\|)^2 \right) \\ &= \sum_{i \neq j} 2w_{ij} (d_{ij} - \|x_i - x_j\|) \nabla_{x_i} (-\|x_i - x_j\|) \\ &= \sum_{i \neq j} 2w_{ij} (d_{ij} - \|x_i - x_j\|) \left(-\frac{x_i - x_j}{\|x_i - x_j\|} \right) \\ &= 2 \sum_{i \neq j} w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|} \right) (x_i - x_j) \end{aligned}$$

Question A.2

Considering a matrix $A \in \mathbb{R}^{k \times k}$ be an orthonormal matrix, $b \in \mathbb{R}^k$ and $1_n \in \mathbb{R}^n$ a vector of all 1's. We want to show that $E(AX + b1_n^T) = E(X)$.

$$\begin{aligned}
E(AX + b1_n^T) &= \sum_{i < j} w_{ij} (d_{ij} - \|Ax_i + b1_n^T - Ax_j - b1_n^T\|)^2 \\
&= \sum_{i < j} w_{ij} (d_{ij} - \|Ax_i - Ax_j\|)^2 \\
&= \sum_{i < j} w_{ij} (d_{ij} - \|A(x_i - x_j)\|)^2 \\
&= \sum_{i < j} w_{ij} (d_{ij} - \|x_i - x_j\|)^2 \\
&= E(X)
\end{aligned}$$

Because the stress $E(X)$ remains identical regardless of how we rotate (A) or shift (b) the entire configuration, the local optima are not unique. Any optimal solution can be transformed into another optimal solution with the same stress value.

We used this property before $\|Ax\| = \|x\|$ with A orthonormal. We want to prove it in order to have a better understanding:

$$\begin{aligned}
\|Ax\|^2 &= (Ax)^T(Ax) \\
&= x^T A^T Ax \\
&= x^T x \\
&= \|x\|^2
\end{aligned}$$

Taking square roots we have proved that $\|Ax\| = \|x\|$

Question A.3

We want to prove that $OPT(k_2) \leq OPT(k_1)$ where $k_1, k_2 \in N$ with $0 < k_1 < k_2$ and the optimal stress in dimension k is defined as $OPT(k) = \inf_{X \in \mathbb{R}^{k \times n}} E(X)$

Let $\mathbf{X}^* \in \mathbb{R}^{k_1 \times n}$ be an optimal solution for dimension k_1 . This means that $\mathbf{X}^*$ minimizes the stress function E over all possible configurations in $\mathbb{R}^{k_1 \times n}$. Therefore:

$$E(\mathbf{X}^*) = OPT(k_1)$$

We denote the columns of $\mathbf{X}^*$ as $\mathbf{x}_1^*, \mathbf{x}_2^*, \dots, \mathbf{x}_n^*$, where each $\mathbf{x}_i^* \in \mathbb{R}^{k_1}$ represents the coordinates of point i in the k_1 -dimensional embedding.

Now we construct a configuration $\tilde{\mathbf{X}} \in \mathbb{R}^{k_2 \times n}$ by embedding $\mathbf{X}^*$ into the higher-dimensional space $\mathbb{R}^{k_2}$. We do this by appending $(k_2 - k_1)$ zero coordinates to each point:

$$\tilde{\mathbf{X}} = \begin{bmatrix} \mathbf{X}^* \\ \mathbf{0} \end{bmatrix} \in \mathbb{R}^{k_2 \times n}$$

where $\mathbf{0} \in \mathbb{R}^{(k_2-k_1) \times n}$ is a matrix containing only zeros.

More explicitly, for each point $i = 1, 2, \dots, n$, we define:

$$\tilde{\mathbf{x}}_i = \begin{bmatrix} \mathbf{x}_i^* \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} x_{i,1}^* \\ x_{i,2}^* \\ \vdots \\ x_{i,k_1}^* \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \in \mathbb{R}^{k_2}$$

where the last $(k_2 - k_1)$ components are all zero.

Now we compute the Euclidean distance between any two points $\tilde{\mathbf{x}}_i$ and $\tilde{\mathbf{x}}_j$ in this higher-dimensional space:

$$\begin{aligned} \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|^2 &= \left\| \begin{bmatrix} \mathbf{x}_i^* \\ \mathbf{0} \end{bmatrix} - \begin{bmatrix} \mathbf{x}_j^* \\ \mathbf{0} \end{bmatrix} \right\|^2 \\ &= \left\| \begin{bmatrix} \mathbf{x}_i^* - \mathbf{x}_j^* \\ \mathbf{0} \end{bmatrix} \right\|^2 \\ &= \sum_{\ell=1}^{k_1} (x_{i,\ell}^* - x_{j,\ell}^*)^2 + \sum_{\ell=k_1+1}^{k_2} 0^2 \\ &= \sum_{\ell=1}^{k_1} (x_{i,\ell}^* - x_{j,\ell}^*)^2 \\ &= \|\mathbf{x}_i^* - \mathbf{x}_j^*\|^2 \end{aligned}$$

Taking square roots on both sides:

$$\|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\| = \|\mathbf{x}_i^* - \mathbf{x}_j^*\|$$

This shows that the Euclidean distances between all pairs of points are preserved under this embedding. The additional zero coordinates do not contribute to the distance calculation.

Now we compute the stress value for the configuration $\tilde{\mathbf{X}}$ in the k_2 -dimensional space. Using the definition of the stress function from Equation (2):

$$E(\tilde{\mathbf{X}}) = \sum_{i < j} w_{ij} (d_{ij} - \|\tilde{\mathbf{x}}_i - \tilde{\mathbf{x}}_j\|)^2$$

Substituting the distance preservation property we just proved:

$$\begin{aligned} E(\tilde{\mathbf{X}}) &= \sum_{i < j} w_{ij} (d_{ij} - \|\mathbf{x}_i^* - \mathbf{x}_j^*\|)^2 \\ &= E(\mathbf{X}^*) \end{aligned}$$

Since $\mathbf{X}^*$ was the optimal configuration in dimension k_1 , we have:

$$E(\tilde{\mathbf{X}}) = E(\mathbf{X}^*) = \text{OPT}(k_1)$$

Now, observe that $\tilde{\mathbf{X}} \in \mathbb{R}^{k_2 \times n}$ is a feasible configuration in the k_2 -dimensional space. By definition, $\text{OPT}(k_2)$ is the minimum value of the stress function over all possible configurations in $\mathbb{R}^{k_2 \times n}$:

$$\text{OPT}(k_2) = \min_{\mathbf{Y} \in \mathbb{R}^{k_2 \times n}} E(\mathbf{Y})$$

Since $\tilde{\mathbf{X}}$ is one particular configuration in $\mathbb{R}^{k_2 \times n}$, and the minimum is taken over all such configurations, we must have:

$$\text{OPT}(k_2) \leq E(\tilde{\mathbf{X}})$$

Combining this with our earlier result:

$$\text{OPT}(k_2) \leq E(\tilde{\mathbf{X}}) = E(\mathbf{X}^*) = \text{OPT}(k_1)$$

Therefore:

$$\text{OPT}(k_2) \leq \text{OPT}(k_1)$$

This proves that the minimum achievable stress in dimension k_2 is at most the minimum achievable stress in dimension k_1 . In other words, as we increase the embedding dimension, the optimal stress value can only decrease or remain the same, but it cannot increase.

The intuition behind this result is that higher-dimensional spaces provide more flexibility and degrees of freedom for positioning the points. Any configuration that exists in a lower-dimensional space can be embedded into a higher-dimensional space (by adding zero coordinates) without changing the pairwise distances. Therefore, the set of achievable distance configurations in higher dimensions is at least as large as in lower dimensions, which means we can achieve at least as good an approximation to the target distances d_{ij} .

Question A.4

We want to explain why a local optima of the stress function satisfy

$$\sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|\mathbf{x}_i - \mathbf{x}_j\|} \right) (\mathbf{x}_i - \mathbf{x}_j) = \mathbf{0}$$

We found the gradient of the stress function in the question A.1, which is:

$$\nabla_{x_i} E = 2 \sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|} \right) (x_i - x_j)$$

A local optima of a differentiable function must satisfy the first-order optimality condition: the gradient must be zero $\nabla_{x_i} E = 0$.

$$\nabla_{x_i} E = 2 \sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|} \right) (x_i - x_j) = 0$$

We divide by the constant 2, in order to get the desired expression

$$\sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|} \right) (x_i - x_j) = 0$$

This represents a balance of forces: each point experiences forces from all other points.

Question A.5

The local optimum condition from Question A.4 can be written as:

$$\sum_{j \neq i} w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|} \right) (x_i - x_j) = 0$$

This represents a **balance of forces** acting on each point x_i . Each neighboring point x_j exerts a force on x_i along the direction $(x_i - x_j)$ with magnitude proportional to $w_{ij} \left(1 - \frac{d_{ij}}{\|x_i - x_j\|} \right)$.

Case (i): $\|x_i - x_j\| > d_{ij}$ (Points are too far apart)

When the embedded distance exceeds the target distance:

$$1 - \frac{d_{ij}}{\|x_i - x_j\|} > 0$$

The force is in the direction $(x_i - x_j)$, which points *away* from x_j . However, since this represents a spring-like attraction, the positive coefficient means point x_j exerts an attractive force on x_i , pulling it closer to reduce the excessive distance.

Interpretation: Points that are too far apart attract each other to decrease their separation.

Case (ii): $\|x_i - x_j\| < d_{ij}$ (Points are too close)

When the embedded distance is less than the target distance:

$$1 - \frac{d_{ij}}{\|x_i - x_j\|} < 0$$

The force has a negative coefficient, reversing the direction. Point x_j now exerts a repulsive force on x_i , pushing it away to increase their separation.

Interpretation: Points that are too close together repel each other to increase their distance.

Equilibrium

At a local optimum, the sum of all attractive and repulsive forces on each point equals zero. This creates a configuration where:

- Pairs with $\|x_i - x_j\| > d_{ij}$ contribute attractive forces
- Pairs with $\|x_i - x_j\| < d_{ij}$ contribute repulsive forces
- The weighted sum of these forces balances out at each point

This mechanical analogy is similar to a spring-mass system where springs have rest lengths d_{ij} and stiffness w_{ij} , and the system reaches equilibrium when all forces balance.

```
import pandas as pd
import torch
import torch.nn.functional as F
from pathlib import Path
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from typing import Tuple, Optional, List, Union
```

```
class MEPSimilarityAnalyzer:
```

```
    """
```

```
    Analyzes voting similarity between Members of European Parliament (MEPs)
    using centered and L2-normalized dot product (cosine similarity).
    Implemented with PyTorch for efficient GPU computation.
```

```
    """
```

```
    def __init__(self, rcv_files: Union[str, List[str]], voted_docs_files: Optional[Union[str, List[str]]]
```

```
        """
        Initialize the analyzer with data files.
```

```
        Args:
```

```
            rcv_files: Path to RCV file (str) or list of paths for multiple sessions
            voted_docs_files: Optional path(s) to voted docs file(s) for metadata
            device: 'cuda' for GPU or 'cpu' for CPU computation
```

```
        """
```

```
        # Convert to list if single file
```

```
        if isinstance(rcv_files, str):
```

```
            self.rcv_files = [rcv_files]
```

```
        ,
```

```

else:
    self.rcv_files = list(rcv_files)

    # Handle voted_docs_files
    if voted_docs_files is None:
        self.voted_docs_files = [None] * len(self.rcv_files)
    elif isinstance(voted_docs_files, str):
        self.voted_docs_files = [voted_docs_files]
    else:
        self.voted_docs_files = list(voted_docs_files)

self.device = torch.device(device if torch.cuda.is_available() else 'cpu')
print(f"Using device: {self.device}")
print(f"Number of files to load: {len(self.rcv_files)}")

self.mep_data = None
self.vote_matrix = None
self.similarity_matrix = None
self.voted_docs = None

def load_data(self) -> pd.DataFrame:
    """
    Load the RCV data from Excel file(s).
    If multiple files, simply stacks all MEPs and all votes together.

    Returns:
        DataFrame containing MEP information and votes
    """
    print(f"\n=== Loading data ===")

    """

```

```

all_dfs = []

for i, rcv_file in enumerate(self.rcv_files):
    print(f"Loading file {i+1}/{len(self.rcv_files)}: {rcv_file}")
    df = pd.read_excel(rcv_file)

    # Add a column to track which file/session this came from
    df['Source_File'] = rcv_file
    df['File_Index'] = i

    print(f" - {len(df)} MEPs, {len(df.columns) - 10} votes")
    all_dfs.append(df)

if len(all_dfs) == 1:
    # Single file
    self.mep_data = all_dfs[0]
else:
    # Multiple files: stack MEPs vertically
    print(f"\nStacking {len(all_dfs)} datasets...")
    self.mep_data = pd.concat(all_dfs, ignore_index=True)

# Store metadata columns (first 10 columns according to codebook)
self.metadata_cols = self.mep_data.columns[:10].tolist()

# Vote columns start from column 11 (index 10)
# Need to find where vote columns end (before Source_File and File_Index)
vote_end_idx = self.mep_data.columns.get_loc('Source_File')
self.vote_cols = self.mep_data.columns[10:vote_end_idx].tolist()

print(f"\n=== Dataset summary ===")
print(f"Total rows (MEPs × Sessions): {len(self.mep_data)}")
print(f"Total vote columns: {len(self.vote_cols)}")
print(f"Unique MEPs (by name): {self.mep_data['FullName'].unique().tolist()}")

```

```

print("\nUnique MEPS (by name): {self.mep_data['File_Index'].unique()}")

if len(self.rcv_files) > 1:
    print("\nBreakdown by file:")
    for i, rcv_file in enumerate(self.rcv_files):
        count = (self.mep_data['File_Index'] == i).sum()
        print(f"  File {i+1}: {count} MEPS")

return self.mep_data

def load_vote_metadata(self):
    """
    Load vote metadata from Voted_docs files including sponsor information.
    """
    print("\n=== Loading vote metadata ===")

    all_docs = []

    for i, docs_file in enumerate(self.voted_docs_files):
        if docs_file is None:
            print(f"Warning: No voted_docs file provided for session {i+1}")
            continue

        print(f"Loading metadata file {i+1}: {docs_file}")

        try:
            df = pd.read_excel(docs_file)
            df['File_Index'] = i
            all_docs.append(df)
            print(f"  - Loaded {len(df)} vote records")
        except Exception as e:
            print(f"  - Error loading file: {e}")

```



```
if len(all_docs) == 0:
    print("No metadata files loaded!")
    return None

# Concatenate all metadata
self.voted_docs = pd.concat(all_docs, ignore_index=True)

print(f"\nTotal vote records: {len(self.voted_docs)}")

# Check for author_name column
if 'Author' in self.voted_docs.columns:
    author_col = 'Author'
elif 'author_name' in self.voted_docs.columns:
    author_col = 'author_name'
else:
    print("Warning: No 'Author' or 'author_name' column found!")
    print(f"Available columns: {self.voted_docs.columns.tolist()}")
    return self.voted_docs

# Count votes by sponsor
sponsor_counts = self.voted_docs[author_col].value_counts()
print(f"\nTop 10 sponsors by number of votes:")
print(sponsor_counts.head(10))

return self.voted_docs

def encode_votes(self, votes_raw: torch.Tensor) -> torch.Tensor:
    """
    Encode votes according to the scheme:
    1 (for) -> +1
    2 (against) -> -1
    3 (abstention) -> 0
    """
```

```

- , .....
0, 4, 5, 6 (not MEP, absent, did not vote, motivated) → NaN (to be handled)

```

Args:

votes_raw: Tensor of raw vote values

Returns:

Tensor of encoded votes with NaN for invalid votes

```

"""

```

```

encoded = torch.full_like(votes_raw, float('nan'), dtype=torch.float32)

```

```

encoded[votes_raw == 1] = 1.0 # For

```

```

encoded[votes_raw == 2] = -1.0 # Against

```

```

encoded[votes_raw == 3] = 0.0

```

```

encoded[votes_raw == 4] = 0.0

```

```

encoded[votes_raw == 5] = 0.0

```

```

encoded[votes_raw == 6] = 0.0

```

```

# 0 remains as NaN

```

```

return encoded

```

```

def preprocess_data(self, min_participation_rate: float = 0.1) -> torch.Tensor:

```

```

"""

```

Preprocess the voting data:

1. Encode votes
2. Filter MEPs by minimum participation rate
3. Center the data (subtract mean for each MEP)
4. L2 normalize each MEP's voting vector

Args:

min_participation_rate: Minimum fraction of votes an MEP must participate in

Returns:

Preprocessed vote matrix (n_meps × n_votes)

```

#####
if self.mep_data is None:
    self.load_data()

print("\n=== Preprocessing votes ===")

# Extract vote columns and convert to PyTorch tensor
votes_raw = torch.tensor(
    self.mep_data[self.vote_cols].values,
    dtype=torch.float32,
    device=self.device
)
n_meps, n_votes = votes_raw.shape

print(f"Initial shape: {n_meps} MEPS × {n_votes} votes")

# Step 1: Encode votes
print("Step 1: Encoding votes (For→+1, Against→-1, Abstain→0)...")
votes_encoded = self.encode_votes(votes_raw)

# Step 2: Filter MEPS by participation rate
print(f"Step 2: Filtering MEPS with participation rate >= {min_participation_rate:.1%}...")
valid_votes_per_mep = (~torch.isnan(votes_encoded)).sum(dim=1)
participation_rate = valid_votes_per_mep / n_votes

active_meps_mask = participation_rate >= min_participation_rate
votes_filtered = votes_encoded[active_meps_mask]
self.mep_data_filtered = self.mep_data[active_meps_mask.cpu().numpy()].reset_index(drop=True)

print(f"Kept {active_meps_mask.sum().item()} MEPS (removed {(~active_meps_mask).sum().item()})")

# Step 3: Center the data (subtract mean for each MEP)

```

```

print("Step 3: Centering data (subtracting each MEP's mean vote)...")

# Create mask for valid votes
valid_mask = ~torch.isnan(votes_filtered)

# Replace NaN with 0 for computation
votes_no_nan = torch.where(valid_mask, votes_filtered, torch.zeros_like(votes_filtered))

# Compute mean for each MEP (only over valid votes)
valid_counts = valid_mask.sum(dim=1, keepdim=True).clamp(min=1) # Avoid division by zero

# Center each vote
mean_votes = votes_no_nan.sum(dim=0, keepdim=True) / valid_counts # Mean across MEPs (dim=0)
votes_centered = votes_no_nan - mean_votes

# Step 4: L2 normalize each MEP's voting vector
print("Step 4: L2 normalizing each MEP's voting vector...")

# Compute L2 norm for each MEP
norms = torch.norm(votes_centered, dim=1, keepdim=True).clamp(min=1e-8) # Avoid division by zero
votes_normalized = votes_centered / norms

self.vote_matrix = votes_normalized

print(f"\nFinal preprocessed matrix shape: {votes_normalized.shape}")
print(f"Data is now centered and L2-normalized")
return votes_normalized

def compute_similarity_matrix(self, batch_size: Optional[int] = None) -> torch.Tensor:
    """
    Compute the pairwise similarity matrix using dot product
    (which is cosine similarity after centering and L2 normalization).

```

Uses batched computation for memory efficiency with large datasets.

Args:

batch_size: If specified, compute similarity in batches (useful for very large datasets)

Returns:

Symmetric similarity matrix (n_meps x n_meps)

.....

if self.vote_matrix is None:

raise ValueError("Must call preprocess_data() first")

print("\n=== Computing similarity matrix ===")

print("Using cosine similarity (dot product of centered + L2-normalized vectors)")

n_meps = self.vote_matrix.shape[0]

if batch_size is None or n_meps <= batch_size:

Compute all at once: $V \times V^T$

self.similarity_matrix = torch.mm(self.vote_matrix, self.vote_matrix.t())

else:

Compute in batches to save memory

print(f"Computing in batches of {batch_size}...")

self.similarity_matrix = torch.zeros(n_meps, n_meps, device=self.device)

for i in range(0, n_meps, batch_size):

end_i = min(i + batch_size, n_meps)

batch = self.vote_matrix[i:end_i]

self.similarity_matrix[i:end_i] = torch.mm(batch, self.vote_matrix.t())

if (i // batch_size + 1) % 10 == 0:

print(f" Processed {end_i}/{n_meps} MEPS")

```

print(f"Computed {n_meps} x {n_meps} similarity matrix")

# Verify symmetry
is_symmetric = torch.allclose(
    self.similarity_matrix,
    self.similarity_matrix.t(),
    atol=1e-5
)
print(f"Matrix is symmetric: {is_symmetric}")

# Verify diagonal values (should be ~1 for normalized vectors)
diag_values = torch.diag(self.similarity_matrix)
print(f"Diagonal values (self-similarity): min={diag_values.min().item():.4f}, "
      f"max={diag_values.max().item():.4f}, mean={diag_values.mean().item():.4f}")

# Statistics (exclude diagonal)
mask = ~torch.eye(n_meps, dtype=torch.bool, device=self.device)
off_diag = self.similarity_matrix[mask]

print(f"\nSimilarity statistics (off-diagonal):")
print(f"  Min: {off_diag.min().item():.4f}")
print(f"  Max: {off_diag.max().item():.4f}")
print(f"  Mean: {off_diag.mean().item():.4f}")
print(f"  Std: {off_diag.std().item():.4f}")

return self.similarity_matrix

def get_most_similar_meps(self, mep_idx: int, top_k: int = 10) -> pd.DataFrame:
    """
    Find the most similar MEPS to a given MEP.

    Args:
        mep_idx: Index of the MEP to find similar ones for.
        top_k: Number of most similar MEPS to return.
    """

```

```

        mep_idx: Index of the MEP in the filtered dataset
        top_k: Number of most similar MEPs to return

Returns:
    DataFrame with most similar MEPs and their similarity scores
    """
    if self.similarity_matrix is None:
        raise ValueError("Must call compute_similarity_matrix() first")

    # Get similarities for this MEP
    similarities = self.similarity_matrix[mep_idx].clone()

    # Exclude self (diagonal)
    similarities[mep_idx] = float('-inf')

    # Get top k indices
    top_values, top_indices = torch.topk(similarities, k=top_k)

    # Convert to CPU for DataFrame creation
    top_values = top_values.cpu().numpy()
    top_indices = top_indices.cpu().numpy()

    # Create result DataFrame
    results = []
    for rank, (idx, sim) in enumerate(zip(top_indices, top_values), 1):
        mep_info = self.mep_data_filtered.iloc[idx]
        results.append({
            'Rank': rank,
            'Name': mep_info['FullName'],
            'Country': mep_info['Country'],
            'Party': mep_info['Party'],
            'EPG': mep_info['EPG'],
            'Similarity': sim
        })

```

```

        'Similarity': sim
    })

    return pd.DataFrame(results)

def analyze_by_group(self, group_col: str = 'EPG') -> pd.DataFrame:
    """
    Analyze average within-group and between-group similarities.

    Args:
        group_col: Column name to group by (e.g., 'EPG', 'Country', 'Party')

    Returns:
        DataFrame with average similarities for each group pair
    """
    if self.similarity_matrix is None:
        raise ValueError("Must call compute_similarity_matrix() first")

    groups = self.mep_data_filtered[group_col].values
    unique_groups = sorted([g for g in set(groups) if pd.notna(g)])

    results = []

    # Move similarity matrix to CPU for numpy indexing
    sim_cpu = self.similarity_matrix.cpu().numpy()

    for g1 in unique_groups:
        mask1 = groups == g1
        for g2 in unique_groups:
            mask2 = groups == g2

            # Get submatrix
            import numpy as np

```



```

import numpy as np
submatrix = sim_cpu[np.ix_(mask1, mask2)]

# If same group, exclude diagonal
if g1 == g2:
    mask = ~np.eye(submatrix.shape[0], dtype=bool)
    avg_sim = submatrix[mask].mean()
else:
    avg_sim = submatrix.mean()

results.append({
    'Group1': g1,
    'Group2': g2,
    'Count1': np.sum(mask1),
    'Count2': np.sum(mask2),
    'Avg_Similarity': avg_sim
})

return pd.DataFrame(results)

def visualize_similarity_matrix(self, figsize=(12, 10), sample_size: Optional[int] = None):
    """
    Visualize the similarity matrix as a heatmap.

    Args:
        figsize: Figure size
        sample_size: If specified, randomly sample this many MEPs for visualization
    """
    if self.similarity_matrix is None:
        raise ValueError("Must call compute_similarity_matrix() first")

    # Move to CPU for visualization
    sim_matrix = self.similarity_matrix.cpu().numpy()

```

```

sim_matrix = self.similarity_matrix_eppa(), numpy(),

# Sample if requested
if sample_size is not None and sample_size < len(sim_matrix):
    import numpy as np
    indices = np.random.choice(len(sim_matrix), sample_size, replace=False)
    sim_matrix = sim_matrix[np.ix_(indices, indices)]
    title_suffix = f" (Random sample of {sample_size} MEPs)"
else:
    title_suffix = ""

plt.figure(figsize=figsize)
sns.heatmap(sim_matrix, cmap='RdBu_r', center=0,
            vmin=-1, vmax=1, square=True,
            cbar_kws={'label': 'Cosine Similarity'})
plt.title(f'MEP Voting Similarity Matrix{title_suffix}')
plt.xlabel('MEP Index')
plt.ylabel('MEP Index')
plt.tight_layout()
plt.show()

def visualize_group_similarity(self, group_col: str = 'EPG', figsize=(10, 8)):
    """
    Visualize average similarity between political groups.

    Args:
        group_col: Column to group by
        figsize: Figure size
    """
    df = self.analyze_by_group(group_col)

    # Create pivot table for heatmap
    pivot = df.pivot(index='Group1', columns='Group2', values='Ava Similarity')

```

```

pivot = pivot_table(index=groups, columns=groups, values='avg_similarity',

plt.figure(figsize=figsize)
sns.heatmap(pivot, annot=True, fmt='.3f', cmap='RdBu_r', center=0,
            vmin=-1, vmax=1, square=True,
            cbar_kws={'label': 'Average Cosine Similarity'})
plt.title(f'Average Voting Similarity by {group_col}')
plt.tight_layout()
plt.show()

def save_similarity_matrix(self, filename: str):
    """Save similarity matrix to disk."""
    torch.save(self.similarity_matrix.cpu(), filename)
    print(f"Similarity matrix saved to '{filename}'")

def load_similarity_matrix(self, filename: str):
    """Load similarity matrix from disk."""
    self.similarity_matrix = torch.load(filename, map_location=self.device)
    print(f"Similarity matrix loaded from '{filename}'")

def compute_mds(self, n_components: int = 2) -> torch.Tensor:
    """
    Compute Multidimensional Scaling (MDS) coordinates from similarity matrix.
    Uses direct eigenvalue decomposition of the similarity matrix S.

    Since S is symmetric positive semi-definite, we have:
     $S = U \Lambda U^T = (\Lambda^{0.5} U^T)^T (\Lambda^{0.5} U^T)$ 

    So we can take  $X = \Lambda^{0.5} U^T$  (selecting top k components)
    such that  $S \approx X^T X$ 

    Args:

```

```

    n_components: Number of dimensions for the embedding (typically 2 for visualization)

Returns:
    Tensor of shape (n_meps, n_components) containing the MDS coordinates
"""
if self.similarity_matrix is None:
    raise ValueError("Must call compute_similarity_matrix() first")

print(f"\n=== Computing MDS with {n_components} components ===")
print("Method: Direct eigenvalue decomposition of similarity matrix")
print("Formula:  $X = I_{k \times n}^{\Lambda^{0.5}} U^T$ ")

S = self.similarity_matrix
n = S.shape[0]

# Step 1: Eigenvalue decomposition of similarity matrix
#  $S = U \Lambda U^T$ 
print("\nComputing eigenvalue decomposition of similarity matrix...")
eigenvalues, eigenvectors = torch.linalg.eigh(S)

# Step 2: Sort eigenvalues in descending order
idx = torch.argsort(eigenvalues, descending=True)
eigenvalues = eigenvalues[idx]
eigenvectors = eigenvectors[:, idx]

print(f"Top 10 eigenvalues: {eigenvalues[:10].cpu().numpy()}")

# Step 3: Select top k components
# Take only non-negative eigenvalues (there might be small negative ones due to numerical errors)
positive_mask = eigenvalues > 1e-10
n_positive = positive_mask.sum().item()

```

```
if n_positive < n_components:
    print(f"Warning: Only {n_positive} positive eigenvalues found, using all of them")
    n_components = n_positive

eigenvalues_k = eigenvalues[:n_components]
eigenvectors_k = eigenvectors[:, :n_components]

# Ensure eigenvalues are non-negative (clamp small negative values to 0)
eigenvalues_k = torch.clamp(eigenvalues_k, min=0)

# Step 4: Compute  $X = \Lambda^{\{0.5\}} U^T$ 
# This gives us X such that  $X^T X \approx S$ 
# Transposed to get (n_meps, n_components) instead of (n_components, n_meps)
Lambda_sqrt = torch.diag(torch.sqrt(eigenvalues_k))
X = (Lambda_sqrt @ eigenvectors_k.T).T # Transpose to get (n, k)

self.mds_coordinates = X.cpu().numpy()

# Compute explained variance
total_variance = eigenvalues[eigenvalues > 0].sum()
explained_variance = eigenvalues_k.sum() / total_variance * 100

print(f"\nMDS complete!")
print(f"Coordinates shape: {self.mds_coordinates.shape}")
print(f"Explained variance by top {n_components} components: {explained_variance.item():.2f}%")
print(f"Number of positive eigenvalues: {n_positive}")

# Verify reconstruction (optional sanity check)
S_reconstructed = X @ X.T
reconstruction_error = torch.norm(S - S_reconstructed, 'fro') / torch.norm(S, 'fro')
print(f"Relative reconstruction error: {reconstruction_error.item():.6f}")
```

```
    return X

def plot_mds(self, color_by: str = 'EPG', figsize=(14, 10), save_path: Optional[str] = None,
             sample_size: Optional[int] = None, alpha: float = 0.6):
    """
    Plot MDS coordinates colored by a specified attribute.

    Args:
        color_by: Column to color points by ('EPG', 'Country', 'Party')
        figsize: Figure size
        save_path: If provided, save figure to this path
        sample_size: If provided, randomly sample this many MEPs for clearer visualization
        alpha: Transparency of points
    """
    if self.mds_coordinates is None:
        raise ValueError("Must call compute_mds() first")

    import numpy as np

    # Get coordinates and labels
    X = self.mds_coordinates
    labels = self.mep_data_filtered[color_by].values

    # Handle sampling if requested
    if sample_size is not None and sample_size < len(X):
        indices = np.random.choice(len(X), sample_size, replace=False)
        X = X[indices]
        labels = labels[indices]
        title_suffix = f" (Random sample of {sample_size} MEPs)"
    else:
        title_suffix = ""
```

```

# Create figure
fig, ax = plt.subplots(figsize=figsize)

# Get unique labels (excluding NaN)
unique_labels = sorted([l for l in set(labels) if pd.notna(l)])

# Create color map
if color_by == 'EPG':
    # Use specific colors for political groups
    color_map = {
        'EPP': '#3399FF',      # Blue (center-right)
        'S&D': '#FF0000',      # Red (center-left)
        'Renew': '#FFDD00',    # Yellow (liberal)
        'Greens/EFA': '#00CC00', # Green
        'ID': '#000066',       # Dark blue (far-right)
        'ECR': '#0066CC',      # Blue (right)
        'GUE/NGL': '#990000',  # Dark red (far-left)
        'NI': '#999999',       # Gray (non-attached)
    }
    # Add any missing groups with random colors
    for label in unique_labels:
        if label not in color_map:
            color_map[label] = plt.cm.tab20(len(color_map) % 20)
else:
    # Use colormap for countries or parties
    cmap = plt.cm.tab20 if len(unique_labels) <= 20 else plt.cm.hsv
    color_map = {label: cmap(i / len(unique_labels)) for i, label in enumerate(unique_labels)}

# Plot each group
for label in unique_labels:
    mask = labels == label
    ax.scatter(X[mask, 0], X[mask, 1],

```

```

        c=[color_map[label]],
        label=label,
        alpha=alpha,
        s=50,
        edgecolors='black',
        linewidths=0.5)

# Formatting
ax.set_xlabel('MDS Dimension 1', fontsize=12)
ax.set_ylabel('MDS Dimension 2', fontsize=12)
ax.set_title(f'MDS Projection of MEPs colored by {color_by}{title_suffix}', fontsize=14, fontweight='bold')
ax.grid(True, alpha=0.3)

# Legend
if len(unique_labels) <= 15:
    ax.legend(loc='center left', bbox_to_anchor=(1, 0.5), fontsize=10)
else:
    print(f"Too many categories ({len(unique_labels)}) for legend. Legend omitted.")

plt.tight_layout()

if save_path:
    plt.savefig(save_path, dpi=300, bbox_inches='tight')
    print(f"Figure saved to {save_path}")

plt.show()

def compare_mds_colorings(self, figsize=(20, 8), sample_size: Optional[int] = None):
    """
    Create side-by-side plots comparing EPG and Country colorings.

    Args:
        figsize: A tuple representing the figure size in inches (width, height).
        sample_size: An optional integer representing the number of samples to use for the plots.
    """

```



```

        figsize: Figure size
        sample_size: If provided, sample this many MEPS
    """
    if self.mds_coordinates is None:
        raise ValueError("Must call compute_mds() first")

    import numpy as np

    # Get coordinates
    X = self.mds_coordinates
    epq_labels = self.mep_data_filtered['EPG'].values
    country_labels = self.mep_data_filtered['Country'].values

    # Handle sampling
    if sample_size is not None and sample_size < len(X):
        indices = np.random.choice(len(X), sample_size, replace=False)
        X = X[indices]
        epq_labels = epq_labels[indices]
        country_labels = country_labels[indices]
        title_suffix = f" (Sample: {sample_size} MEPS)"
    else:
        title_suffix = ""

    fig, (ax1, ax2) = plt.subplots(1, 2, figsize=figsize)

    # Plot 1: Colored by EPG
    unique_epq = sorted([l for l in set(epq_labels) if pd.notna(l)])
    epq_color_map = {
        'EPP': '#3399FF', 'S&D': '#FF0000', 'Renew': '#FFDD00',
        'Greens/EFA': '#00CC00', 'ID': '#000066', 'ECR': '#0066CC',
        'GUE/NGL': '#990000', 'NI': '#999999',
    }
    for label in unique_epq:

```

```

for label in unique_epg:
    if label not in epg_color_map:
        epg_color_map[label] = plt.cm.tab20(len(epg_color_map) % 20)

for label in unique_epg:
    mask = epg_labels == label
    ax1.scatter(X[mask, 0], X[mask, 1],
                c=[epg_color_map[label]],
                label=label, alpha=0.6, s=50,
                edgecolors='black', linewidths=0.5)

ax1.set_xlabel('MDS Dimension 1', fontsize=11)
ax1.set_ylabel('MDS Dimension 2', fontsize=11)
ax1.set_title(f'Colored by Political Group (EPG){title_suffix}', fontsize=12, fontweight='bold')
ax1.grid(True, alpha=0.3)
ax1.legend(loc='best', fontsize=9)

# Plot 2: Colored by Country
unique_countries = sorted([l for l in set(country_labels) if pd.notna(l)])
cmap = plt.cm.tab20 if len(unique_countries) <= 20 else plt.cm.hsv
country_color_map = {label: cmap(i / len(unique_countries))
                     for i, label in enumerate(unique_countries)}

for label in unique_countries:
    mask = country_labels == label
    ax2.scatter(X[mask, 0], X[mask, 1],
                c=[country_color_map[label]],
                label=label, alpha=0.6, s=50,
                edgecolors='black', linewidths=0.5)

ax2.set_xlabel('MDS Dimension 1', fontsize=11)
ax2.set_ylabel('MDS Dimension 2', fontsize=11)
ax2.set_title(f'Colored by Country{title_suffix}', fontsize=12, fontweight='bold')

```

```

ax2.set_title('Colored by country{title_suffix} , fontsize=12, fontweight= bold ,
ax2.grid(True, alpha=0.3)

if len(unique_countries) <= 15:
    ax2.legend(loc='best', fontsize=8, ncol=2)

plt.tight_layout()
plt.show()

def compute_support_matrix(self, sponsor_col: str = 'Author', min_votes: int = 10):
    """
    Compute support matrix: how each political group votes on proposals
    sponsored by each political group.

    Args:
        sponsor_col: Column name for sponsor ('Author' or 'author_name')
        min_votes: Minimum number of votes a sponsor must have to be included

    Returns:
        DataFrame with support rates (rows=sponsors, cols=voter groups)
    """
    if self.voted_docs is None:
        self.load_vote_metadata()

    if self.voted_docs is None:
        raise ValueError("No vote metadata available")

    if self.mep_data_filtered is None:
        raise ValueError("Must call preprocess_data() first")

    print(f"\n=== Computing Support Matrix ===")
    print(f"Sponsor column: {sponsor_col}")

```

```

# Create mapping from various names to standard short names
epg_name_mapping = {
    'Confederal Group of the European United Left – Nordic Green Left': 'GUE/NGL',
    'EUL/NGL': 'GUE/NGL',
    'The Left': 'GUE/NGL',
    'European Conservatives and Reformists Group': 'ECR',
    'ECR': 'ECR',
    'Group of the European People's Party (Christian Democrats)': 'EPP',
    'EPP': 'EPP',
    'Europe of Freedom and Direct Democracy Group': 'EFDD',
    'EFDD': 'EFDD',
    'Europe of Nations and Freedom Group': 'ENF',
    'ENF': 'ENF',
    'IDG': 'ID',
    'ID': 'ID',
    'Greens/EFA': 'Greens/EFA',
    'Greens/ EFA': 'Greens/EFA',
    'Group of the Greens/European Free Alliance': 'Greens/EFA',
    'Group of the Alliance of Liberals and Democrats for Europe': 'ALDE/Renew',
    'ALDE': 'ALDE/Renew',
    'Renew': 'ALDE/Renew',
    'REG': 'ALDE/Renew',
    'Group of the Progressive Alliance of Socialists and Democrats in the European Parliament': 'S&D',
    'S&D': 'S&D',
    'Non-attached Members': 'NI',
    'NI': 'NI',
}

# Map voter group names to standard names
voter_groups_original = sorted([g for g in self.mep_data_filtered['EPG'].unique() if pd.notna(g)])
voter_groups = sorted(list(set([epg_name_mapping.get(g, g) for g in voter_groups_original])))

```

```
print(f"Voter groups (standardized): {voter_groups}")

# Initialize support matrix
support_data = []

# Determine vote ID column name
if 'Vote ID' in self.voted_docs.columns:
    vote_id_col = 'Vote ID'
elif 'euro_act_id' in self.voted_docs.columns:
    vote_id_col = 'euro_act_id'
else:
    print(f"Available columns in voted_docs: {self.voted_docs.columns.tolist()}")
    raise ValueError("No 'Vote ID' or 'euro_act_id' column found in voted_docs")

print(f"Using vote ID column: {vote_id_col}")

# Get sponsors with enough votes
sponsors = self.voted_docs[sponsor_col].value_counts()
sponsors = sponsors[sponsors >= min_votes].index.tolist()

# Map sponsor names and filter for EPG sponsors
sponsors_mapped = [(s, epq_name_mapping.get(s, s)) for s in sponsors]
sponsors_filtered = [(orig, mapped) for orig, mapped in sponsors_mapped
                       if mapped in voter_groups]

print(f"Analyzing {len(sponsors_filtered)} EPG sponsors with >= {min_votes} votes")

# Sort voted_docs by Vote ID to match column order in RCV files
voted_docs_sorted = self.voted_docs.sort_values(vote_id_col).reset_index(drop=True)

print(f"Total votes in metadata: {len(voted_docs_sorted)}")
print(f"Total vote columns in data: {len(self.vote_cols)}")
```

```
# Match by position
for sponsor_orig, sponsor_std in sponsors_filtered:
    # Get vote records sponsored by this group
    sponsor_mask = voted_docs_sorted[sponsor_col] == sponsor_orig
    sponsor_vote_indices = sponsor_mask[sponsor_mask].index.tolist()

    if len(sponsor_vote_indices) == 0:
        continue

    # These indices correspond to column positions in vote_cols
    valid_indices = [i for i in sponsor_vote_indices if i < len(self.vote_cols)]

    if len(valid_indices) == 0:
        print(f" Warning: No valid vote indices for sponsor {sponsor_std}")
        continue

    print(f" {sponsor_std}: {len(valid_indices)} votes")

    # Get vote column names for these positions
    vote_columns = [self.vote_cols[i] for i in valid_indices]

    # Get votes for these proposals from filtered MEPs
    votes_for_sponsor = self.mep_data_filtered[vote_columns].values

    # Compute support rate for each EPG (using standardized names)
    row_data = {'Sponsor': sponsor_std, 'N_Votes': len(valid_indices)}

    for voter_group_std in voter_groups:
        # Get all original names that map to this standard name
        matching_original_names = [orig for orig in voter_groups_original
                                   if epg_name_mapping.get(orig, orig) == voter_group_std]
```

```
# Get MEPs from any of these matching groups
epg_mask = self.mep_data_filtered['EPG'].isin(matching_original_names)

if epg_mask.sum() == 0:
    row_data[voter_group_std] = np.nan
    continue

epg_votes = votes_for_sponsor[epg_mask]

# Count For votes (encoded as 1)
n_for = np.sum(epg_votes == 1)
n_against = np.sum(epg_votes == 2)
n_abstain = np.sum(epg_votes == 3)
n_total = n_for + n_against + n_abstain

support_rate = n_for / n_total if n_total > 0 else np.nan
row_data[voter_group_std] = support_rate

support_data.append(row_data)

if len(support_data) == 0:
    raise ValueError("No support data computed.")

support_df = pd.DataFrame(support_data)
support_df = support_df.set_index('Sponsor')

print(f"\nSupport matrix computed: {support_df.shape}")
print(f"Sponsors: {len(support_df)}, Voter groups: {len(voter_groups)}")

self.support_matrix = support_df
return support_df
```

```
def visualize_support_heatmap(self, figsize=(12, 10), annot=True):
    """
    Visualize support matrix as a heatmap.

    Args:
        figsize: Figure size
        annot: Whether to annotate cells with values
    """
    if not hasattr(self, 'support_matrix') or self.support_matrix is None:
        raise ValueError("Must call compute_support_matrix() first")

    # Remove N_Votes column for visualization
    plot_data = self.support_matrix.drop('N_Votes', axis=1, errors='ignore')

    plt.figure(figsize=figsize)
    sns.heatmap(plot_data, annot=annot, fmt='.2f', cmap='RdYlGn',
                vmin=0, vmax=1, center=0.5,
                cbar_kws={'label': 'Support Rate (% voting For)'})
    plt.title('Political Group Support Matrix\n(Rows: Sponsors, Columns: Voter Groups)',
              fontsize=14, fontweight='bold')
    plt.xlabel('Voter Group (EPG)', fontsize=12)
    plt.ylabel('Sponsor Group', fontsize=12)
    plt.tight_layout()
    plt.show()

def analyze_group_loyalty_and_alliances(self):
    """
    Analyze in-group loyalty and cross-group alliances.

    Returns:
        Dictionary with loyalty scores and alliance patterns
    """
```



```

"""
if not hasattr(self, 'support_matrix') or self.support_matrix is None:
    raise ValueError("Must call compute_support_matrix() first")

print("\n=== Group Loyalty and Alliance Analysis ===")

support_df = self.support_matrix.drop('N_Votes', axis=1, errors='ignore')

# Convert to numpy array for easier indexing
import numpy as np
support_array = support_df.values
row_labels = support_df.index.tolist()
col_labels = support_df.columns.tolist()

results = {
    'in_group_loyalty': {},
    'strongest_alliances': [],
    'strongest_oppositions': [],
    'asymmetric_alliances': []
}

# 1. In-group loyalty (diagonal)
print("\n1. In-Group Loyalty (do MEPs support their own group's proposals?):")
for i, row_group in enumerate(row_labels):
    if row_group in col_labels:
        j = col_labels.index(row_group)
        loyalty = support_array[i, j]
        if not np.isnan(loyalty):
            results['in_group_loyalty'][row_group] = float(loyalty)
            print(f"  {row_group}: {loyalty:.1%}")

# 2. Strongest cross-group alliances

```

```

print("\n2. Strongest Cross-Group Alliances (support > 60%):")
for i, sponsor in enumerate(row_labels):
    for j, voter in enumerate(col_labels):
        if sponsor != voter:
            support = support_array[i, j]
            if not np.isnan(support) and support > 0.6:
                results['strongest_alliances'].append({
                    'Sponsor': sponsor,
                    'Supporter': voter,
                    'Support_Rate': float(support)
                })
            print(f" {voter} supports {sponsor}: {support:.1%}")

# 3. Strongest oppositions (support < 30%)
print("\n3. Strongest Oppositions (support < 30%):")
for i, sponsor in enumerate(row_labels):
    for j, voter in enumerate(col_labels):
        if sponsor != voter:
            support = support_array[i, j]
            if not np.isnan(support) and support < 0.3:
                results['strongest_oppositions'].append({
                    'Sponsor': sponsor,
                    'Opponent': voter,
                    'Support_Rate': float(support)
                })
            print(f" {voter} opposes {sponsor}: {support:.1%}")

# 4. Asymmetric alliances (A supports B more than B supports A)
print("\n4. Asymmetric Alliances (difference > 20%):")
checked_pairs = set()

for i, group1 in enumerate(row_labels):
    if group1 not in checked_pairs:

```

```

if group1 not in col_labels:
    continue
j1 = col_labels.index(group1)

for k, group2 in enumerate(col_labels):
    if group2 not in row_labels or group1 == group2:
        continue

    # Avoid checking same pair twice
    pair = tuple(sorted([group1, group2]))
    if pair in checked_pairs:
        continue
    checked_pairs.add(pair)

    i2 = row_labels.index(group2)

    # group1 votes on group2's proposals: row=group2, col=group1
    support_1_to_2 = support_array[i2, j1]
    # group2 votes on group1's proposals: row=group1, col=group2
    support_2_to_1 = support_array[i, k]

    if not np.isnan(support_1_to_2) and not np.isnan(support_2_to_1):
        asymmetry = abs(support_1_to_2 - support_2_to_1)
        if asymmetry > 0.2:
            results['asymmetric_alliances'].append({
                'Group1': group1,
                'Group2': group2,
                'G1_supports_G2': float(support_1_to_2),
                'G2_supports_G1': float(support_2_to_1),
                'Asymmetry': float(asymmetry)
            })
            print(f" {group1} → {group2}: {support_1_to_2:.1%}, "
                  f"{group2} → {group1}: {support_2_to_1:.1%} ")

```

```

    if group2 > group1: support_c2_c0_1 = 1%r
    f"( $\Delta$  = {asymmetry:.1%})")

```

```

return results

```

```

# Step 1: Initialize with BOTH rcv_files AND voted_docs_files
analyzer_multi = MEPSimilarityAnalyzer(
    rcv_files=[
        'EP-voting-data/EP8_RCVs_2019_06_25.xlsx',
        'EP-voting-data/EP9_RCVs_2022_06_22.xlsx'
    ],
    voted_docs_files=[
        'EP-voting-data/EP8_Voted_docs.xlsx',
        'EP-voting-data/EP9_Voted_docs.xlsx'
    ],
    device='cpu'
)

analyzer_multi.load_data()
analyzer_multi.preprocess_data(min_participation_rate=0.3) # Lower threshold since combined
analyzer_multi.compute_similarity_matrix()

print("\n=== Political group cohesion across EP8 and EP9 ===")
group_sim = analyzer_multi.analyze_by_group('EPG')
within_group = group_sim[group_sim['Group1'] == group_sim['Group2']]
print(within_group[['Group1', 'Count1', 'Avg_Similarity']].to_string(index=False))

```

```

Using device: cpu
Number of files to load: 2

```

```

=== Loading data ===

```

```
Loading file 1/2: EP-voting-data/EP8_RCVs_2019_06_25.xlsx
- 858 MEPs, 10253 votes
Loading file 2/2: EP-voting-data/EP9_RCVs_2022_06_22.xlsx
- 811 MEPs, 13461 votes
```

```
Stacking 2 datasets...
```

```
=== Dataset summary ===
Total rows (MEPs × Sessions): 1669
Total vote columns: 10251
Unique MEPs (by name): 811
```

```
Breakdown by file:
```

```
File 1: 858 MEPs
File 2: 811 MEPs
```

```
=== Preprocessing votes ===
```

```
Initial shape: 1669 MEPs × 10251 votes
```

```
Step 1: Encoding votes (For→+1, Against→-1, Abstain→0)...
```

```
Step 2: Filtering MEPs with participation rate >= 30.0%...
```

```
Kept 1520 MEPs (removed 149)
```

```
Step 3: Centering data (subtracting each MEP's mean vote)...
```

```
Step 4: L2 normalizing each MEP's voting vector...
```

```
Final preprocessed matrix shape: torch.Size([1520, 10251])
```

```
Data is now centered and L2-normalized
```

```
=== Computing similarity matrix ===
```

```
Using cosine similarity (dot product of centered + L2-normalized vectors)
```

```
Computed 1520 × 1520 similarity matrix
```

```
Matrix is symmetric: True
```

```
Diagonal values (self-similarity): min=1.0000, max=1.0000, mean=1.0000
```

```
Similarity statistics (off-diagonal):
```

```
Min: -0.6136
```

Max: 1.0000
Mean: 0.1164
Std: 0.2855

=== Political group cohesion across EP8 and EP9 ===

	Group1	Count1	Avg_S
Confederal Group of the European United Left – Nordic Green Left		51	
	ECR	64	
	EPP	179	
Europe of Freedom and Direct Democracy Group		42	
Europe of Nations and Freedom Group		35	
European Conservatives and Reformists Group		73	
	Greens/EFA	72	
Group of the Alliance of Liberals and Democrats for Europe		67	
Group of the European People's Party (Christian Democrats)		215	
Group of the Greens/European Free Alliance		50	
Group of the Progressive Alliance of Socialists and Democrats in the European Parliament		187	
	IDG	65	
	...	...	

```
print("\n=== Political group cohesion across EP8 and EP9 ===")
group_sim = analyzer_multi.analyze_by_group('EPG')
within_group = group_sim[group_sim['Group1'] == group_sim['Group2']]
print(within_group[['Group1', 'Count1', 'Avg_Similarity']].to_string(index=False))
```

```
=== Political group cohesion across EP8 and EP9 ===
```

	Group1	Count1	Avg_Sim
Confederal Group of the European United Left – Nordic Green Left		51	(
	ECR	64	(
	EPP	179	(
Europe of Freedom and Direct Democracy Group		42	(
Europe of Nations and Freedom Group		35	(
European Conservatives and Reformists Group		73	(
	Greens/EFA	72	(
Group of the Alliance of Liberals and Democrats for Europe		67	(
Group of the European People's Party (Christian Democrats)		215	(
Group of the Greens/European Free Alliance		50	(
Group of the Progressive Alliance of Socialists and Democrats in the European Parliament		187	(
	IDG	65	(
	NI	42	(
	Non-attached Members	23	(
	REG	106	(
	S&D	146	(
	The Left	39	(

```
# Compute MDS coordinates
analyzer_multi.compute_mds(n_components=2)

# Visualizations
print("\n" + "="*70)
print("Creating visualizations...")
```

```

print("="*70)

# Plot 1: Colored by Political Group
print("\n1. Plotting MEPs colored by Political Group (EPG)...")
analyzer_multi.plot_mds(color_by='EPG', sample_size=None) # Use all MEPs

# Plot 2: Colored by Country
print("\n2. Plotting MEPs colored by Country...")
analyzer_multi.plot_mds(color_by='Country', sample_size=None)

# Plot 3: Side-by-side comparison
print("\n3. Creating side-by-side comparison...")
analyzer_multi.compare_mds_colorings(sample_size=None)

```

```

=== Computing MDS with 2 components ===
Method: Direct eigenvalue decomposition of similarity matrix
Formula:  $X = I_{\{k \times n\}}^{\wedge\{0.5\}} U^T$ 

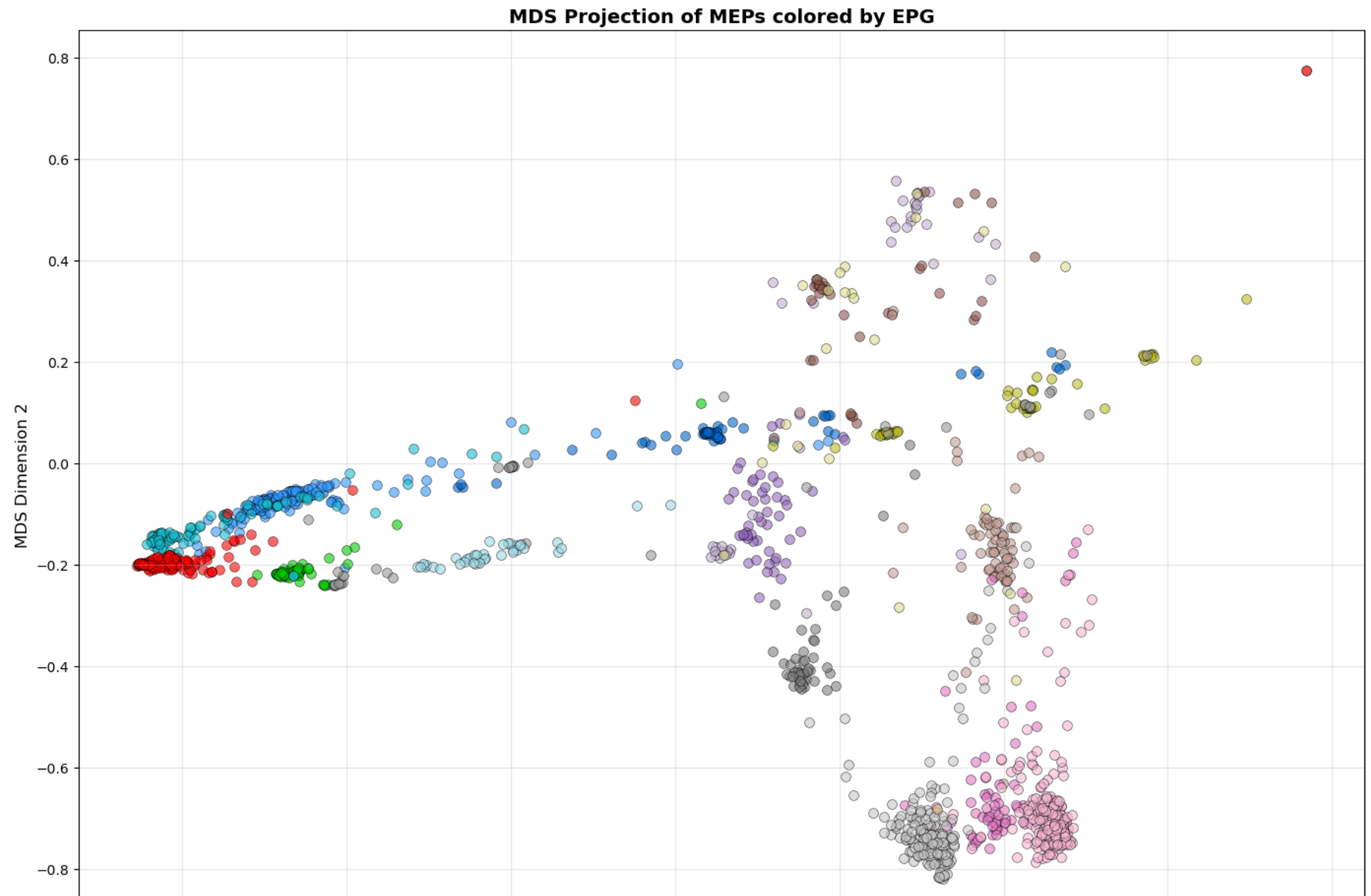
Computing eigenvalue decomposition of similarity matrix...
Top 10 eigenvalues: [316.58914  278.53534  145.30173  121.349014  40.567554  33.707355
 31.091452  27.1568    21.842615  20.3453   ]

MDS complete!
Coordinates shape: (1520, 2)
Explained variance by top 2 components: 39.15%
Number of positive eigenvalues: 1520
Relative reconstruction error: 0.442204

=====
Creating visualizations...
=====

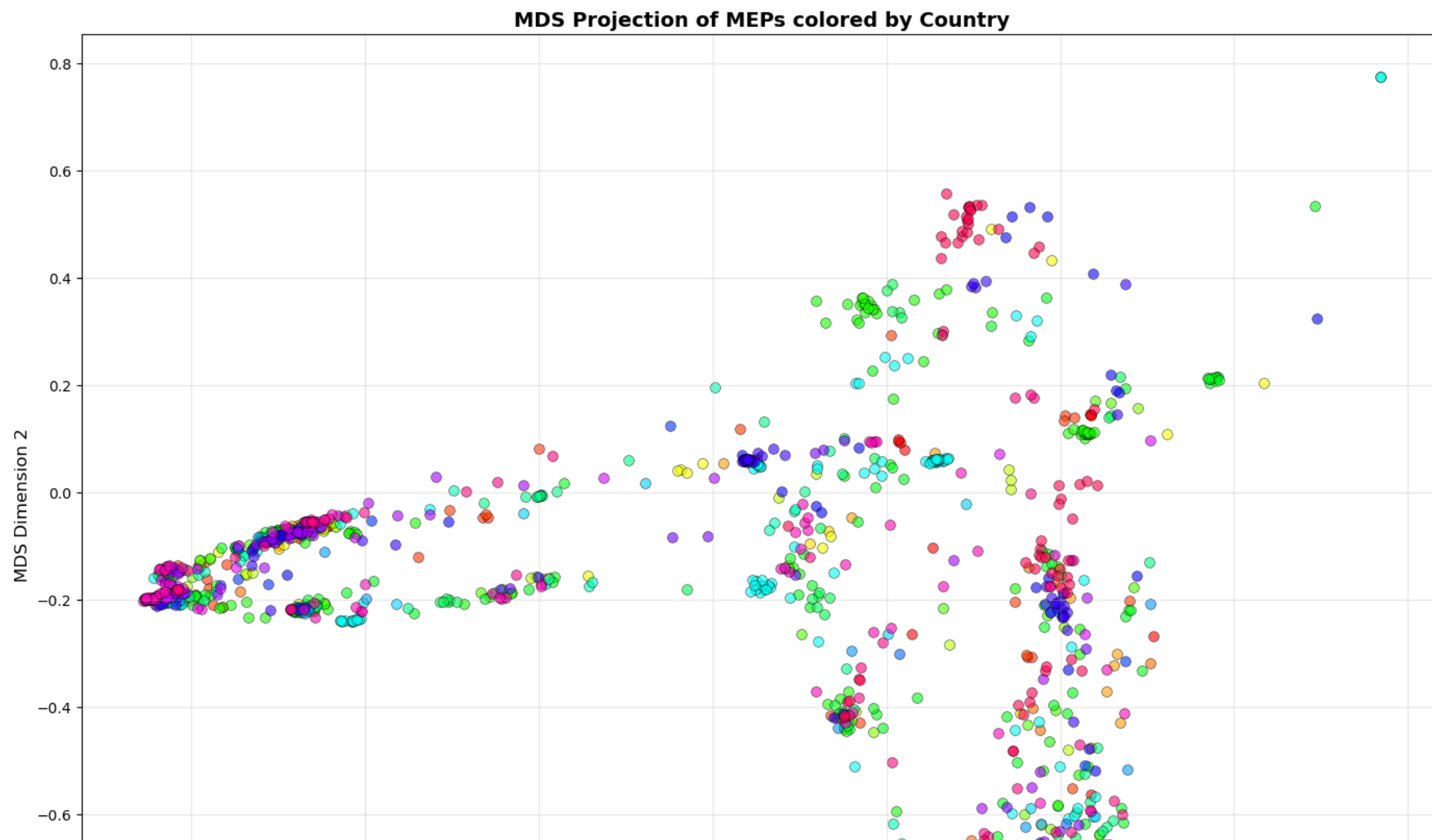
```

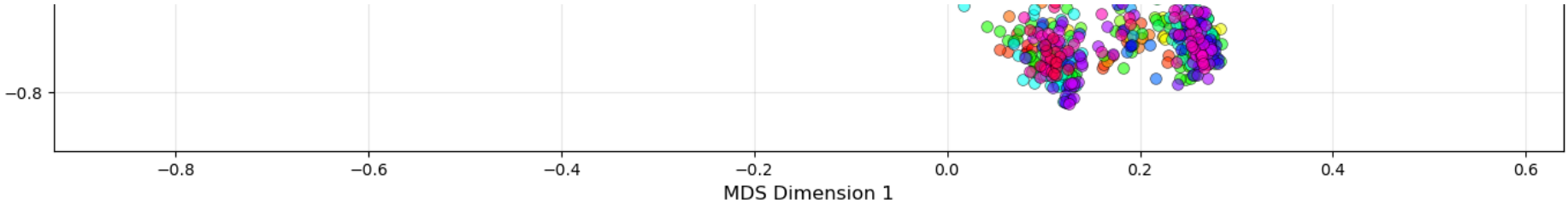

1. Plotting MEPS colored by Political Group (EPG)...
Too many categories (17) for legend. Legend omitted.



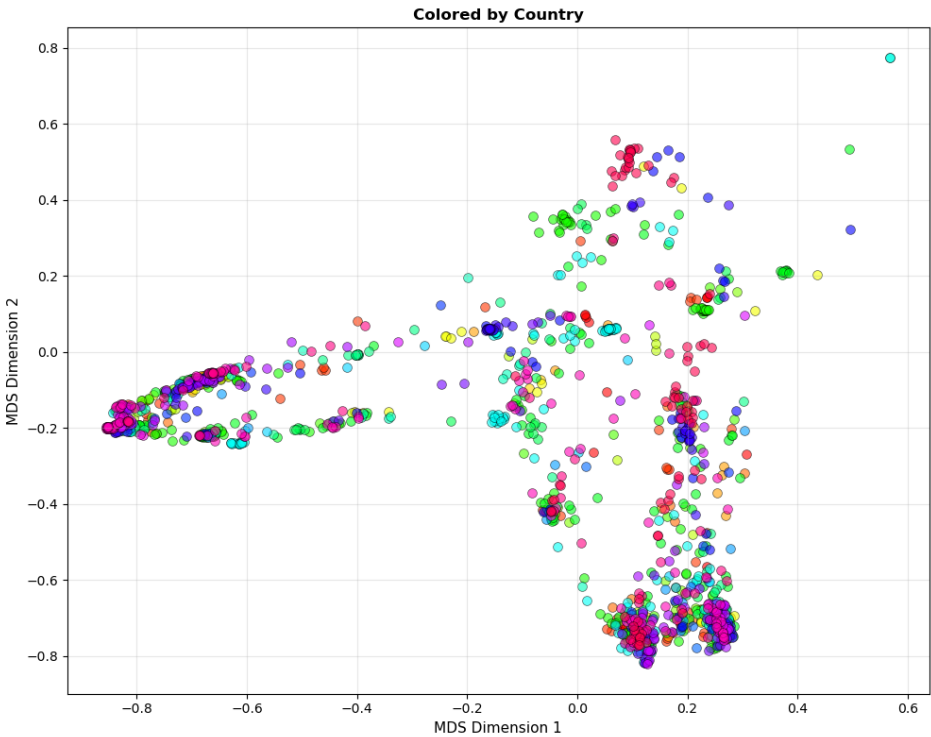
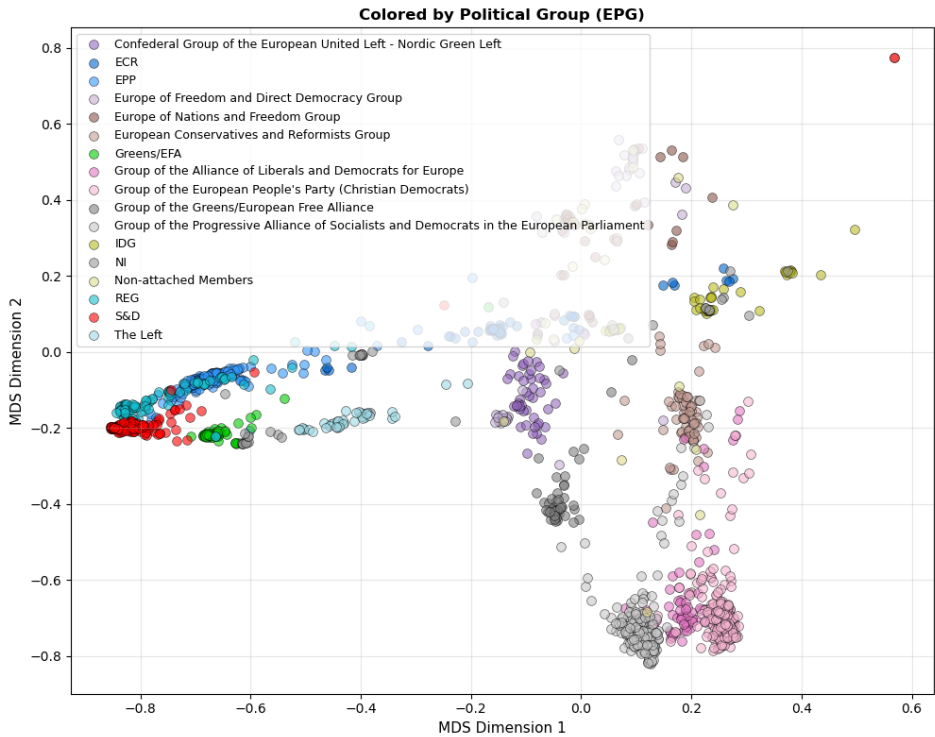
2. Plotting MEPs colored by Country...

Too many categories (28) for legend. Legend omitted.





3. Creating side-by-side comparison...




```
# Analyze just EP9
analyzer_ep9 = MEPSimilarityAnalyzer(
    rcv_files=['EP-voting-data/EP9_RCVs_2022_06_22.xlsx'],
    voted_docs_files=['EP-voting-data/EP9_Voted_docs.xlsx'],
    device='cpu'
)

analyzer_ep9.load_data()
analyzer_ep9.preprocess_data(min_participation_rate=0.5)
analyzer_ep9.load_vote_metadata()
analyzer_ep9.compute_support_matrix(sponsor_col='Author', min_votes=10)
analyzer_ep9.visualize_support_heatmap()
analyzer_ep9.analyze_group_loyalty_and_alliances()
```

Using device: cpu

Number of files to load: 1

=== Loading data ===

Loading file 1/1: EP-voting-data/EP9 RCVs 2022 06 22.xlsx

```

- 811 MEPs, 13461 votes

=== Dataset summary ===
Total rows (MEPs × Sessions): 811
Total vote columns: 13459
Unique MEPs (by name): 811

=== Preprocessing votes ===
Initial shape: 811 MEPs × 13459 votes
Step 1: Encoding votes (For→+1, Against→-1, Abstain→0)...
Step 2: Filtering MEPs with participation rate >= 50.0%...
Kept 705 MEPs (removed 106)
Step 3: Centering data (subtracting each MEP's mean vote)...
Step 4: L2 normalizing each MEP's voting vector...

Final preprocessed matrix shape: torch.Size([705, 13459])
Data is now centered and L2-normalized

=== Loading vote metadata ===
Loading metadata file 1: EP-voting-data/EP9_Voted_docs.xlsx
/opt/anaconda3/lib/python3.12/site-packages/openpyxl/worksheet/_read_only.py:85: UserWarning: Data Validation not supported for this cell
  for idx, row in parser.parse():
  - Loaded 13459 vote records

Total vote records: 13459

Top 10 sponsors by number of votes:
Author
original text      4728
0                  1397
committee          1000
ID                 904
The Left           895
EUL/NGL            760
ECR                728
FPP                542

```

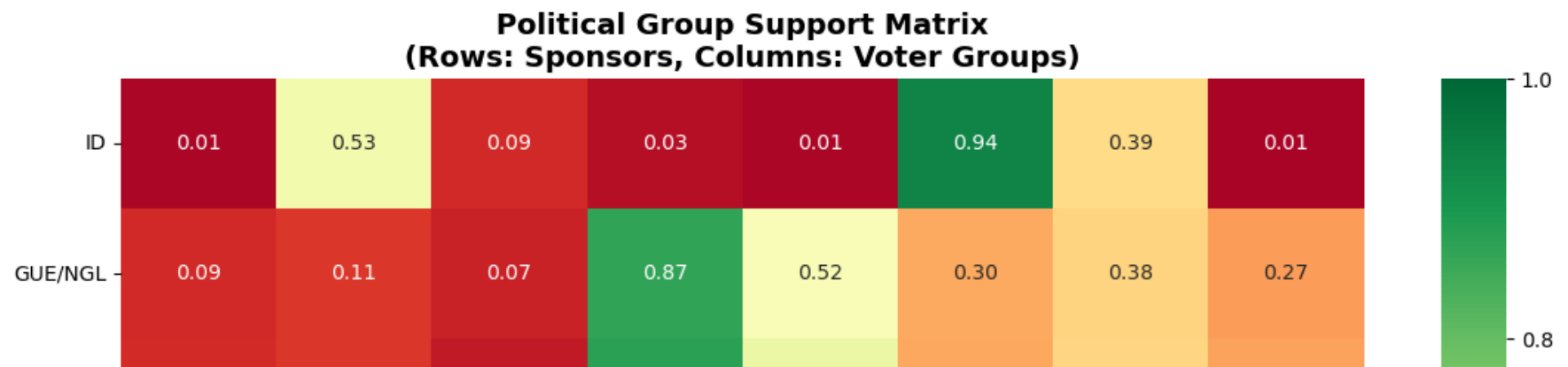
```

---
Greens/ EFA      455
S&D              297
Name: count, dtype: int64

=== Computing Support Matrix ===
Sponsor column: Author
Voter groups (standardized): ['ALDE/Renew', 'ECR', 'EPP', 'GUE/NGL', 'Greens/EFA', 'ID', 'NI', 'S&D']
Using vote ID column: Vote ID
Analyzing 10 EPG sponsors with >= 10 votes
Total votes in metadata: 13459
Total vote columns in data: 13459
  ID: 904 votes
  GUE/NGL: 895 votes
  GUE/NGL: 760 votes
  ECR: 728 votes
  EPP: 542 votes
  Greens/EFA: 455 votes
  S&D: 297 votes
  ALDE/Renew: 185 votes
  Greens/EFA: 155 votes
  GUE/NGL: 19 votes

Support matrix computed: (10, 9)
Sponsors: 10, Voter groups: 8

```





=== Group Loyalty and Alliance Analysis ===

1. In-Group Loyalty (do MEPs support their own group's proposals?):

ID: 93.8%
GUE/NGL: 87.0%
GUE/NGL: 87.8%
ECR: 87.8%
EPP: 93.9%
Greens/EFA: 98.7%
S&D: 97.1%
ALDE/Renew: 90.1%
Greens/EFA: 98.7%
GUE/NGL: 92.3%

2. Strongest Cross-Group Alliances (support > 60%):

ID supports ECR: 69.3%
ECR supports EPP: 74.2%
GUE/NGL supports Greens/EFA: 87.1%
ALDE/Renew supports S&D: 60.8%
GUE/NGL supports S&D: 68.6%
Greens/EFA supports S&D: 74.1%
EPP supports ALDE/Renew: 64.3%
Greens/EFA supports ALDE/Renew: 71.7%
S&D supports ALDE/Renew: 61.8%
GUE/NGL supports Greens/EFA: 90.6%
S&D supports Greens/EFA: 61.1%

3. Strongest Oppositions (support < 30%):

ALDE/Renew opposes ID: 1.2%
EPP opposes ID: 8.8%
GUE/NGL opposes ID: 3.4%
Greens/EFA opposes ID: 1.2%
S&D opposes ID: 0.9%
ALDE/Renew opposes GUE/NGL: 9.0%
ECR opposes GUE/NGL: 11.2%
EPP opposes GUE/NGL: 7.2%
ID opposes GUE/NGL: 29.6%

S&D opposes GUE/NGL: 27.4%
 ALDE/Renew opposes GUE/NGL: 9.3%
 ECR opposes GUE/NGL: 11.3%
 EPP opposes GUE/NGL: 5.7%
 ID opposes GUE/NGL: 29.1%
 S&D opposes GUE/NGL: 28.4%
 ALDE/Renew opposes ECR: 16.3%
 GUE/NGL opposes ECR: 6.4%
 Greens/EFA opposes ECR: 14.8%
 S&D opposes ECR: 11.8%
 GUE/NGL opposes EPP: 19.8%
 Greens/EFA opposes EPP: 25.3%
 S&D opposes EPP: 29.7%
 ALDE/Renew opposes Greens/EFA: 25.5%
 ECR opposes Greens/EFA: 14.5%
 EPP opposes Greens/EFA: 13.6%
 ID opposes Greens/EFA: 20.3%
 ID opposes S&D: 28.3%
 ECR opposes Greens/EFA: 18.0%
 EPP opposes Greens/EFA: 16.3%
 ID opposes Greens/EFA: 19.7%
 ALDE/Renew opposes GUE/NGL: 5.5%
 ECR opposes GUE/NGL: 23.3%
 EPP opposes GUE/NGL: 2.1%
 S&D opposes GUE/NGL: 7.6%

4. Asymmetric Alliances (difference > 20%):

ID → ALDE/Renew: 34.6%, ALDE/Renew → ID: 1.2% (Δ = 33.4%)
 ID → EPP: 53.4%, EPP → ID: 8.8% (Δ = 44.6%)
 ID → GUE/NGL: 29.6%, GUE/NGL → ID: 3.4% (Δ = 26.2%)
 ID → S&D: 28.3%, S&D → ID: 0.9% (Δ = 27.4%)
 GUE/NGL → ALDE/Renew: 57.5%, ALDE/Renew → GUE/NGL: 9.0% (Δ = 48.5%)
 GUE/NGL → Greens/EFA: 87.1%, Greens/EFA → GUE/NGL: 51.7% (Δ = 35.5%)
 GUE/NGL → S&D: 68.6%, S&D → GUE/NGL: 27.4% (Δ = 41.2%)
 ECR → ALDE/Renew: 44.9%, ALDE/Renew → ECR: 16.3% (Δ = 28.6%)
 ECR → EPP: 74.2%, EPP → ECR: 39.5% (Δ = 34.6%)

```

EPP → ALDE/Renew: 64.3%, ALDE/Renew → EPP: 43.8% ( $\Delta$  = 20.5%)
Greens/EFA → ALDE/Renew: 71.7%, ALDE/Renew → Greens/EFA: 25.5% ( $\Delta$  = 46.3%)
Greens/EFA → S&D: 74.1%, S&D → Greens/EFA: 44.5% ( $\Delta$  = 29.6%)
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```

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    'G2_supports_G1': 0.08832701941447978,
    'Asymmetry': 0.44587654484984557},
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```

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```

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